



WINIGEN MATERIALS

Representative Certificate of Analysis

LiBF₄, battery grade

Representative production-lot analytical data

Document No. COA-LiBF4		Issue Date June 19, 2026	Disposition PASS
Product Name	LiBF4	CAS No.	14283-07-9
Lot No.	RC2603007	Batch Quantity	50 kg
Production Date	March 18, 2026	Expiry Date	September 14, 2026
Shelf Life	180 days	Storage Condition	Sealed, dry
Handling	Protect from moisture		

Analytical Results

Test Item	Method	Acceptance Specification	Result	Status
Cadmium (Cd), ppm	ICP-OES	<= 10	<LOQ	PASS
Barium (Ba), ppm	ICP-OES	<= 10	<LOQ	PASS
Manganese (Mn), ppm	ICP-OES	<= 10	<LOQ	PASS
Aluminum (Al), ppm	ICP-OES	<= 10	<LOQ	PASS
Nickel (Ni), ppm	ICP-OES	<= 10	0.11	PASS
Chromium (Cr), ppm	ICP-OES	<= 10	0.02	PASS
Magnesium (Mg), ppm	ICP-OES	<= 10	0.39	PASS
Zinc (Zn), ppm	ICP-OES	<= 10	<LOQ	PASS
Copper (Cu), ppm	ICP-OES	<= 10	<LOQ	PASS
Iron (Fe), ppm	ICP-OES	<= 10	0.99	PASS
Lead (Pb), ppm	ICP-OES	<= 10	<LOQ	PASS
Potassium (K), ppm	ICP-OES	<= 10	0.21	PASS
Calcium (Ca), ppm	ICP-OES	<= 10	1.72	PASS
Sodium (Na), ppm	ICP-OES	<= 10	0.23	PASS
Sulfate, ppm	Turbidimetry	<= 10	< 10	PASS
Chloride (Cl ⁻), ppm	Turbidimetry	<= 5	2.29	PASS
Acidity (as HF), ppm	Acid-base titration	<= 100	15.9	PASS
Purity, wt%	Assay method	>= 99.7	99.758	PASS
Moisture, ppm	Karl Fischer titration	<= 100	15.6	PASS
Color of specified solution, Hazen/Pt-Co	Pt-Co colorimetry; solution per SOP	<= 20	9.9	PASS
Appearance	Visual inspection	White powder, free of visible foreign matter	White powder, free of visible foreign matter	PASS

Quality Review

Analysis Date	June 19, 2026	Approval Date	June 19, 2026
---------------	---------------	---------------	---------------

This representative certificate presents analytical results from a production lot tested against the listed acceptance specifications. Results shown as <LOQ are below the applicable reporting limit. The values shown apply only to the referenced lot/batch and should not be interpreted as results for future production lots.